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ML in Finance Operations

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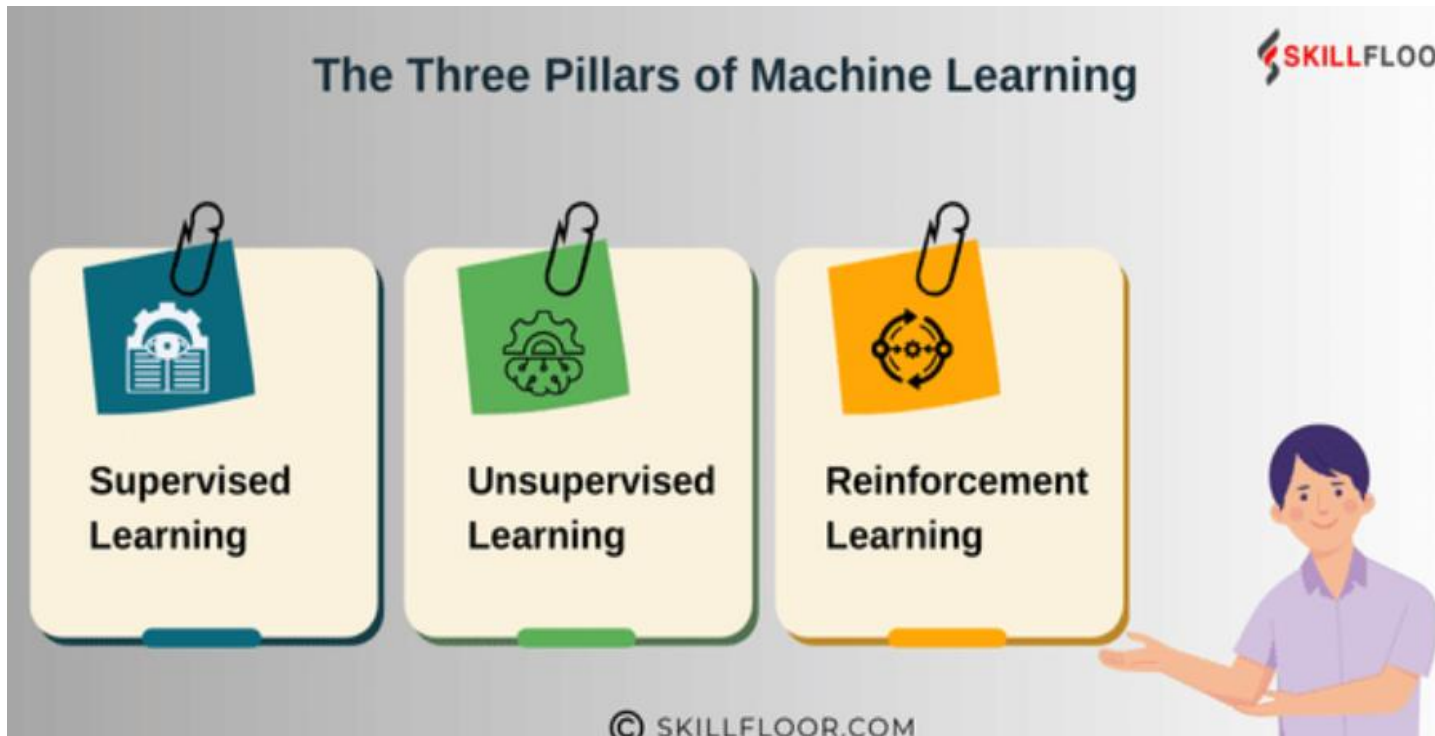
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Outline

- Types of Machine Learning
- Supervised learning methods
 - Logistic Regression, Naive Bayes and Decision trees
 - Evaluation metrics
- Use cases of Supervised learning methods

Types of ML used in Finance

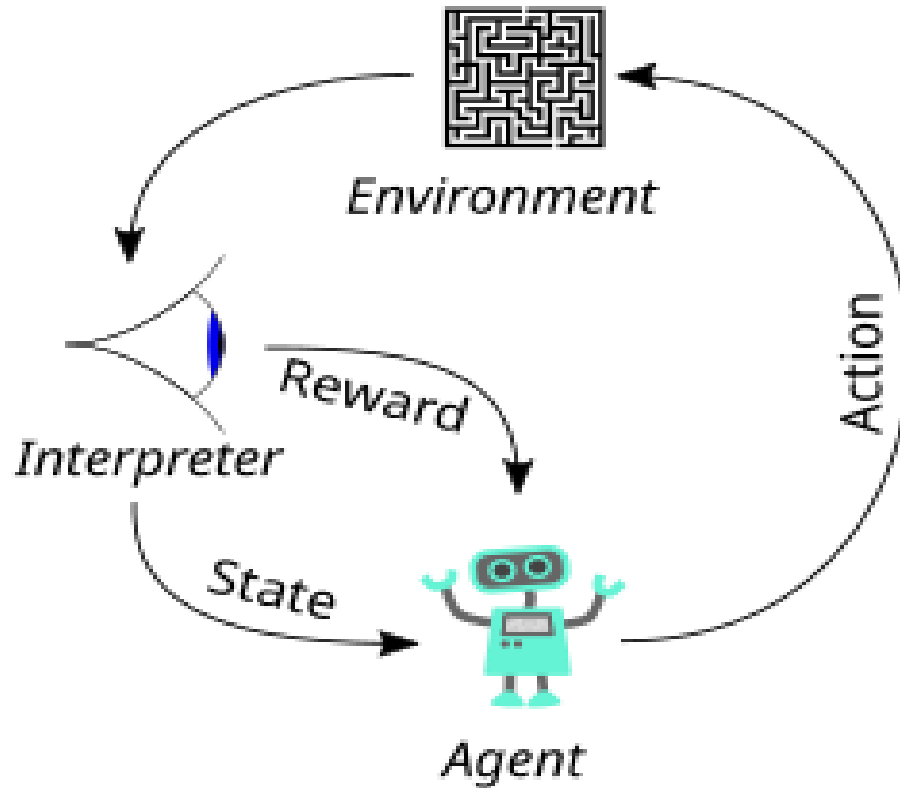


Key Points of ML Types

- Supervised Learning:
 - Learn from **labels** (a template of true/ false), (yes/no) to make conjectures about new data/ future
- Unsupervised Learning:
 - Learn from **patterns** in the data.
- Reinforcement learning
 - Learn from experiences and reward system in the environment

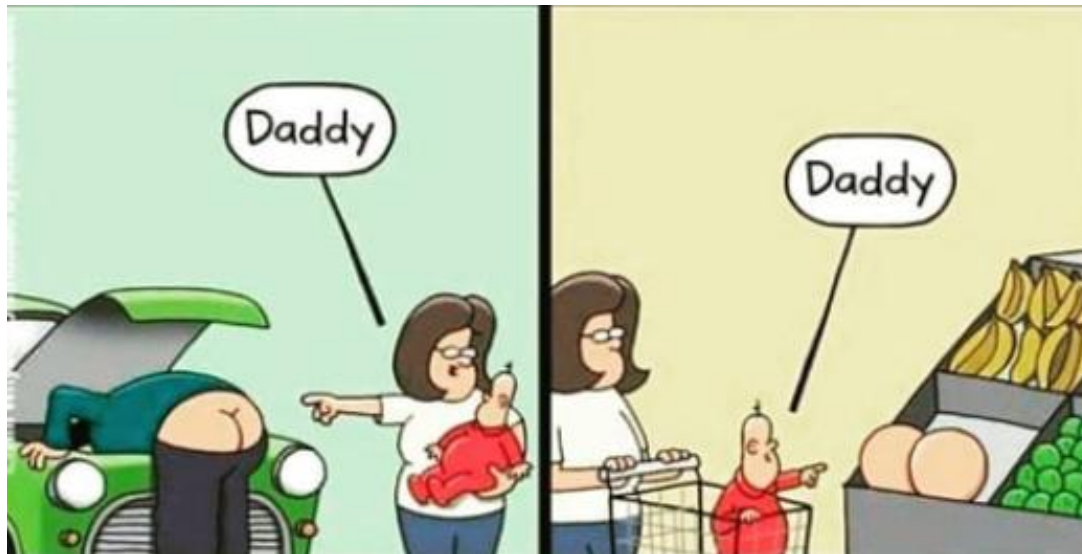
Reinforcement Learning

- Reinforcement learning
 - How agents learn from the environment and a reward system



Supervised Learning

- Supervised learning is a form of ML that use *labelled* data to maps input data to the outcome.
- Classification & regression are the primary supervised methods.



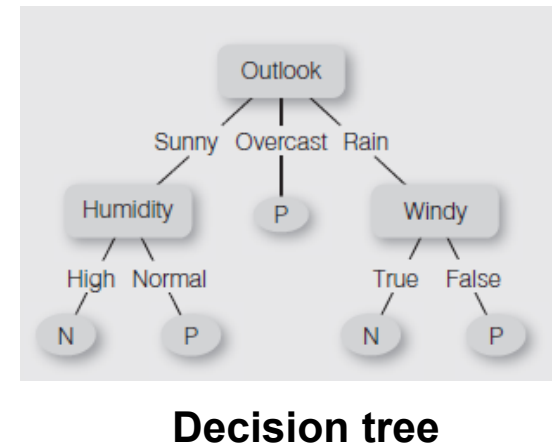
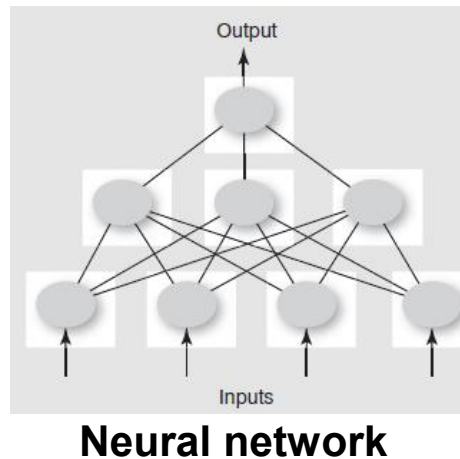
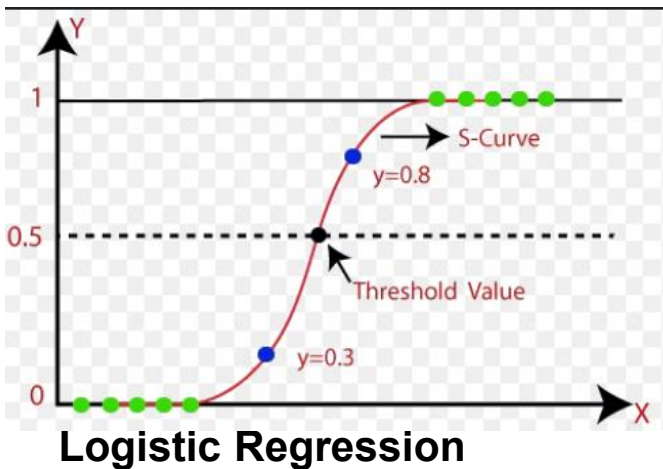
Some classification models

- Logistic regression
- Naïve Bayes
- Decision tree
- Ensemble (meta) models

- Neural networks (including deep learning)
- Support vector machines (SVM)

Classification Models

- Egs of classification models

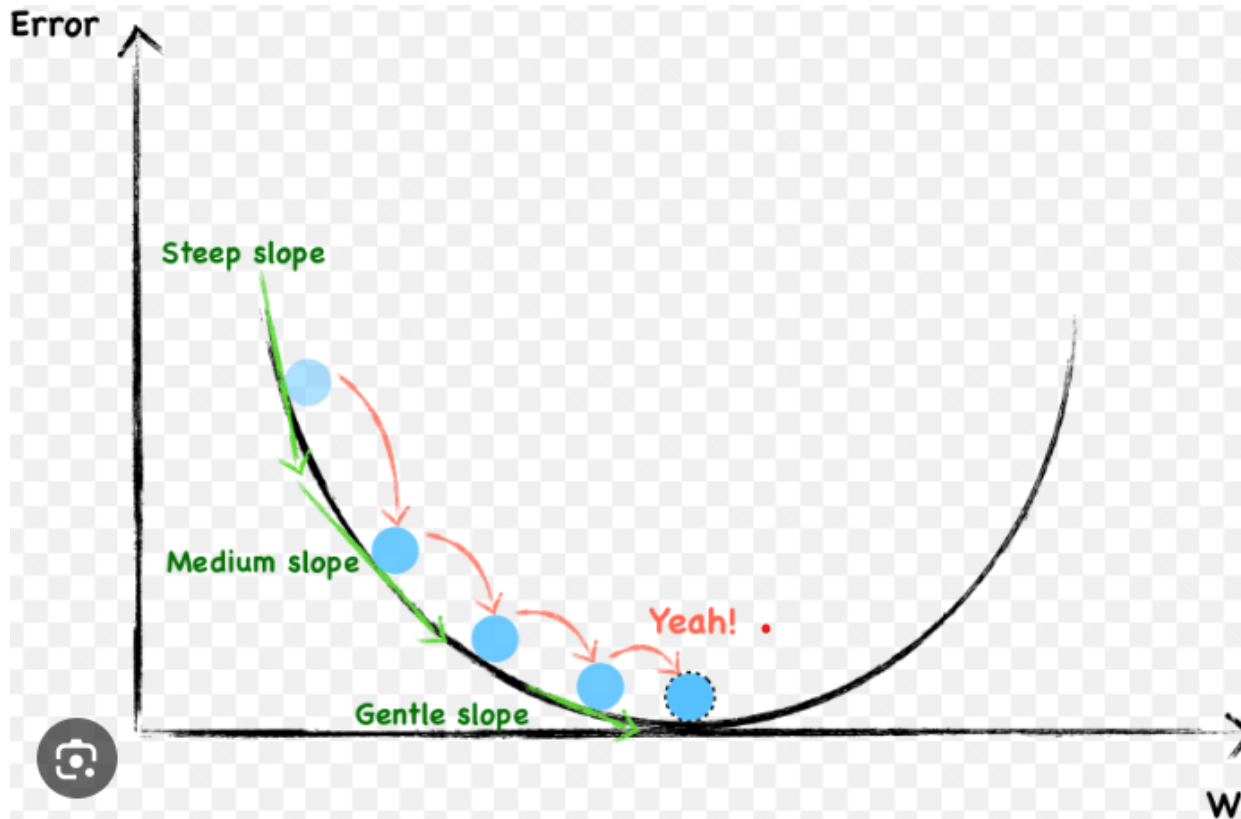


1. Outlook = overcast $\rightarrow$ class = P
2. Outlook = sunny and Humidity = high $\rightarrow$ class = N
3. Outlook = sunny $\rightarrow$ class = P
4. Windy = true $\rightarrow$ class = N
5. () $\rightarrow$ class = P

List of ordered classification rules

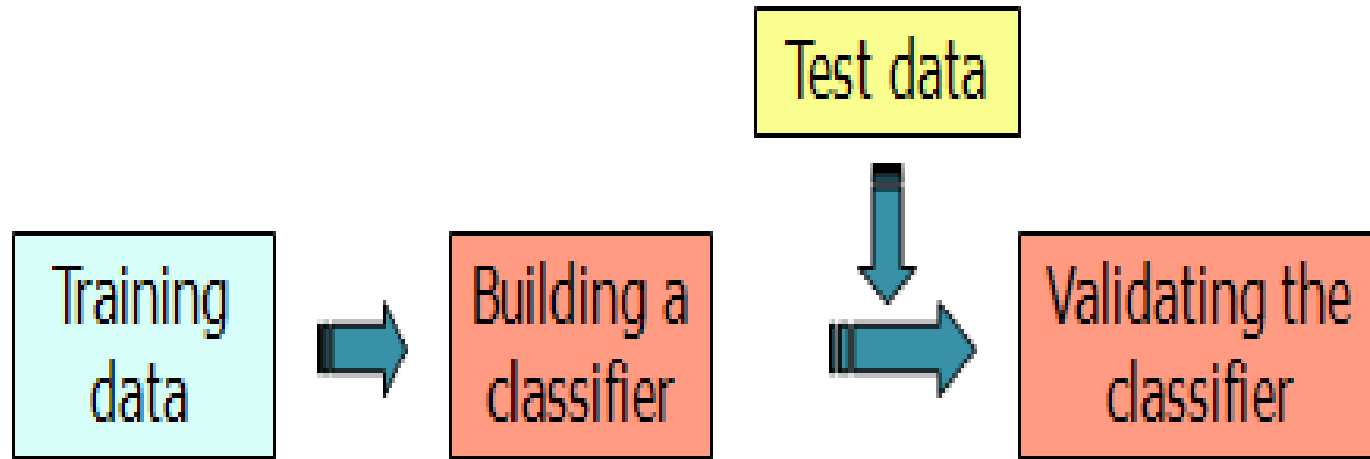
Generally, each method works by **minimising prediction errors** in the training dataset by adjusting the **parameters**

How Errors are reduced in training?



- Method called gradient descent. Parameters are adjusted to reduce the errors 'going down'

Workflow of Classification

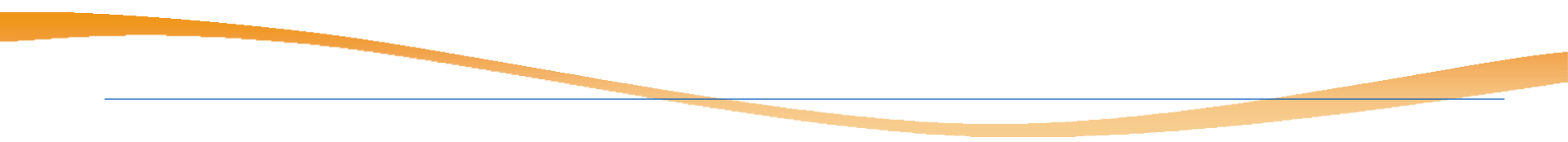


Steps to do Classification

- **What is needed? DATA**
 - Labelled Datasets = training + test sets
- **How does it work? TRAINING**
 - Set up some rules (model classifier) to predict (classify) unseen data
- **Are the rules valid? TESTING**
 - Test those rules with existing data with established metrics

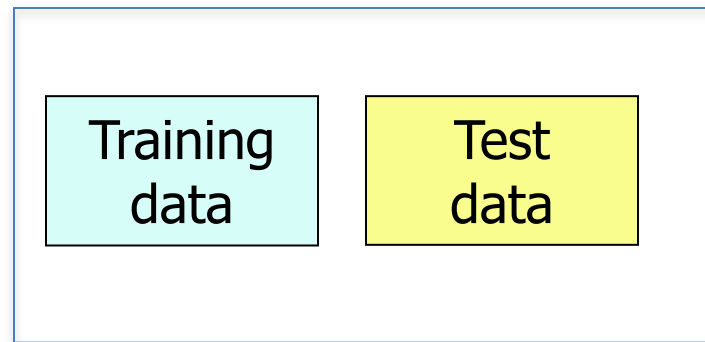
Steps to do Classification: Dividing Data

- Divide the total dataset into 2 (or 3) sets:
 - **Training data** is used for learning the parameters of the model.
 - **Test data** is used to get a final, unbiased estimate of how well the classifier works.
 - This estimate is usually worse than on the training data or validation data.
 - In production environments, sometimes a **validation data set** is used. However, that is optional, especially if data is scarce.



Data Partition

- There are different ways to split your data into a training set / test set.
 - Resubstitution
 - Hold-out
 - Cross-validation
 - Bootstrap



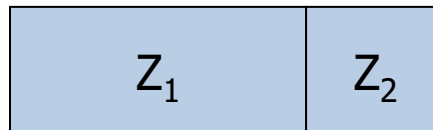
Data Partition: Re-Substitution & Hold-out

1. Re-substitution

- Use training data as test data. Only used if lack of data.

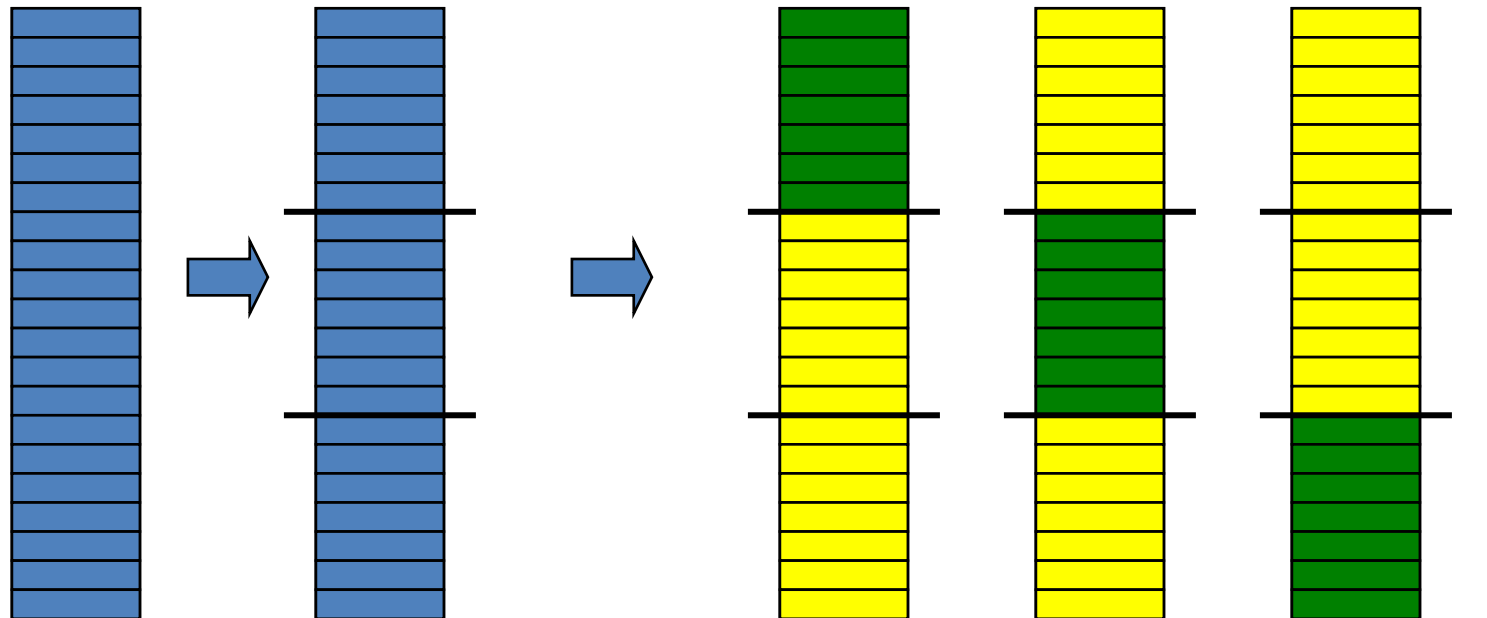
2. Hold-out

- Split data into two parts.
- Use **the larger part (~80%)** for training and **the other** for testing (~20%)
- Most common method



Data Partition: n-fold cross-validation

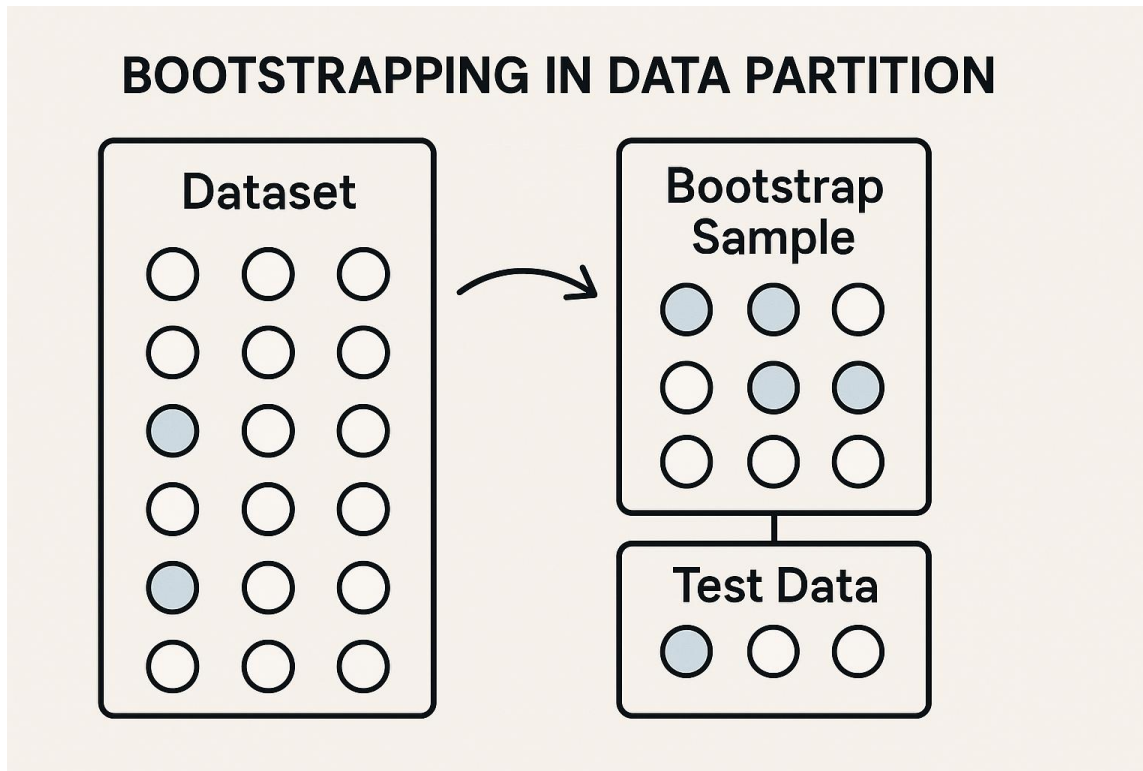
- Divide data into n equal parts. Sequentially, use $n-1$ for training and one part for testing. Average out the test results.
- To resolve uneven data granularity



Training data 
Test data 

evaluate statistics for each iteration
and then compute the average

Data Partition: Bootstrapping



- Data is randomly sampled (from same pool) repeatedly for training/ test.
- Test results are averaged out again.

Common Issues in Classification

- Representativeness of the data set
 - Hopefully, test set is representative for operational data
- Scarce data: major problem
 - Data is expensive, or scarce in Finance.
 - You cannot do trial experiments in finance
- Common mistakes
 - Use Learning (Training) data as Test data
 - What's called **re-substitution** accuracy (sometimes unavoidable)

Confusion Matrix (2x2)

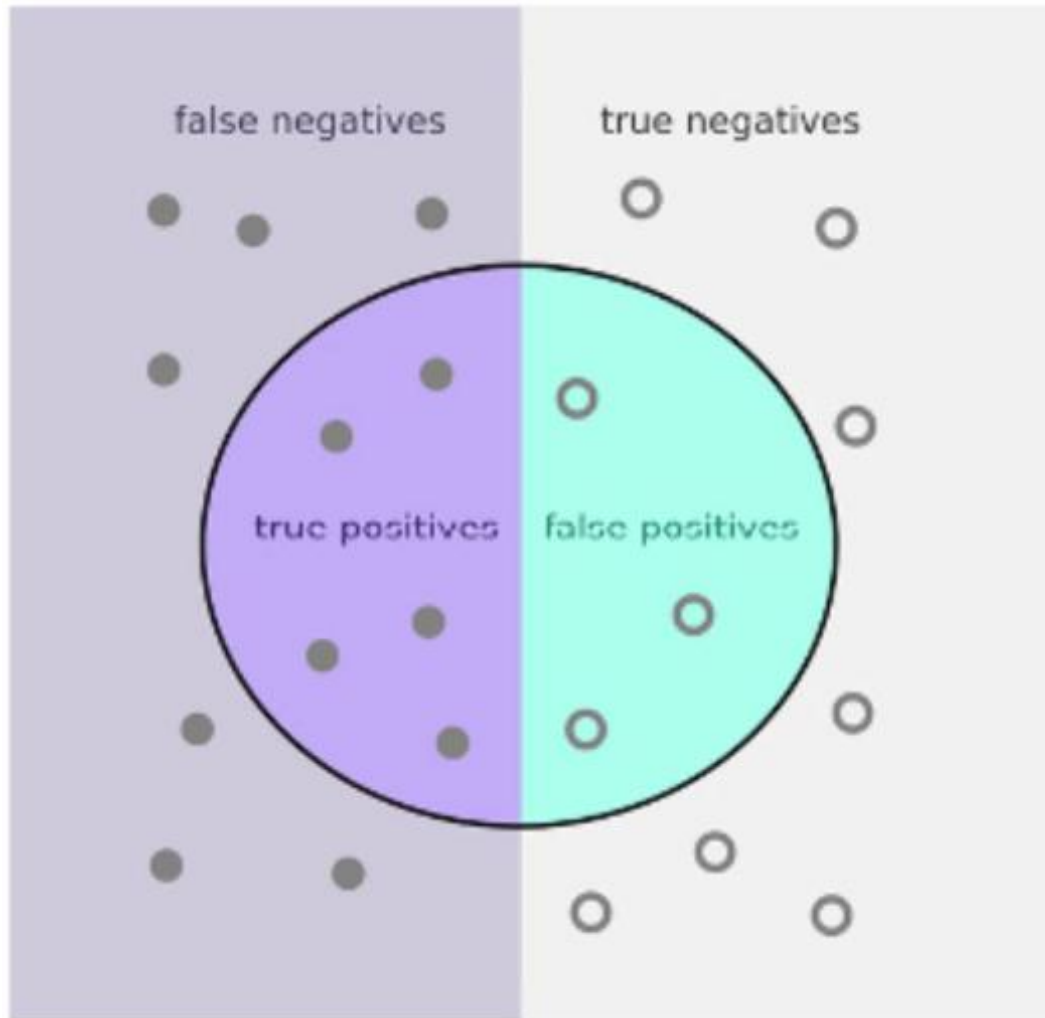
		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Lists out all the possible contingencies for model output. Also called contingency table

- Accuracy : Proportion of correctly classified examples.

- $$Accuracy = \frac{TP+TN}{TP+FP+FN+TN}$$

Types of Errors



- 2 types of errors that exist:
- You guessed Positive but it is actually Negative
 - You guessed Negative it is actually Positive

Recall/ Sensitivity/ True Positive Rate

- Recall/ Sensitivity/ TPR =
 - **TP / (TP + FN)**
 - Proportion of correctly detected positive examples
- Sensitivity/ recall is an important criterion when the infection rate is low. (very few positive).
- The test needs to be v. 'sensitive' (cautious) in order to pick up the few positively infected.

Actual class

	Yes	No
Predicted class	Yes	No
	TP	FP
	FN	TN

Contingency table (2X2)

Precision

- Precision = $TP / (TP + FP)$
 - proportion of cases predicted as positive are indeed positive. What proportion of predicted positive cases are actually positive?
- Use in marketing, stock price prediction

Actual class

	Yes	No
Predicted class	Yes	No
	TP	FP
	FN	TN

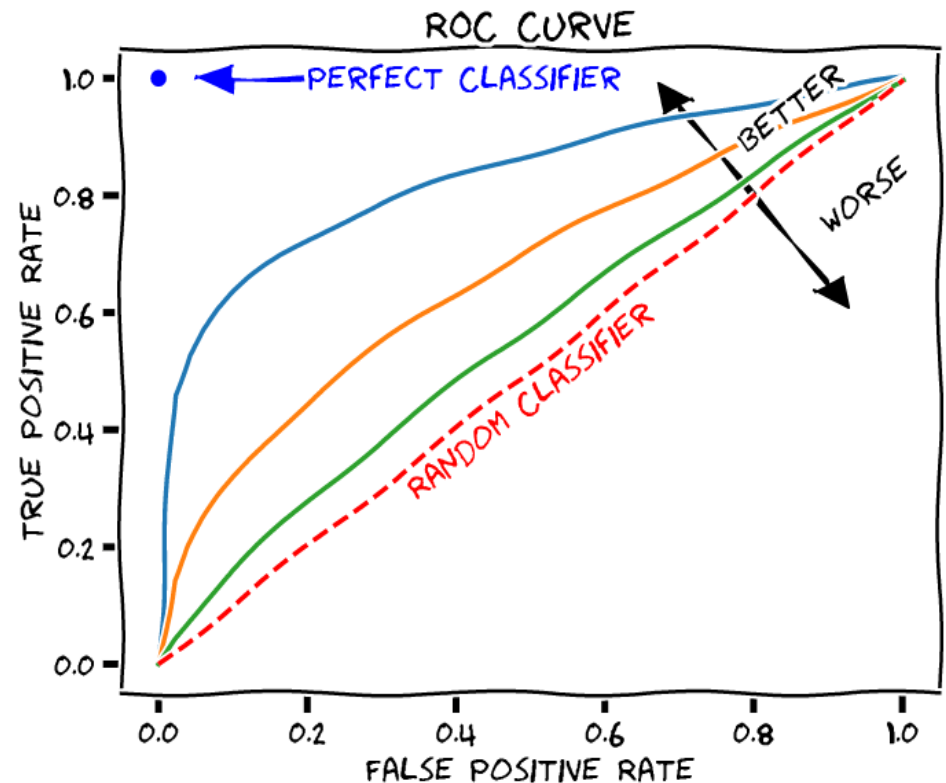
Contingency table (2X2)

No free lunch in ML

- No single algorithm is universally best for all problems.
 - You cannot minimise both error types simultaneously—reducing one usually increases the other.
- Due to threshold tuning (stringency)
 - Lower threshold → catch more defaults (fewer false negatives) but more false alarms (more false positives).
 - Higher threshold → fewer false alarms but more misses.
- The “best” trade-off depends on the economic cost function (e.g., in lending, missing a default is usually much costlier than wrongly rejecting a safe borrower).

Receiver Operating Characteristic (ROC)

- Plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various classification thresholds. A common summary metric for a ROC curve is the Area Under the Curve (AUC). Best = 1.
- Method to determine the 'overall effectiveness' of the model (considering trade-off).



Key Issue on Classification (ML in general)

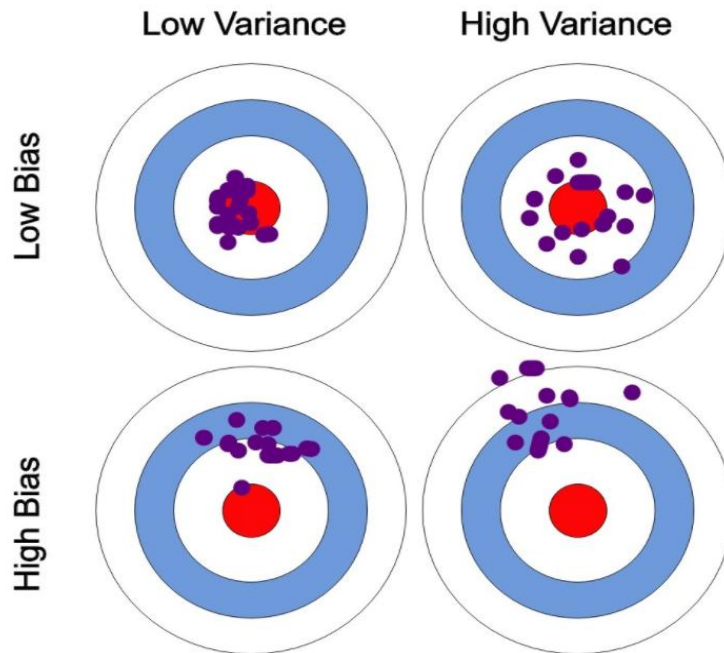
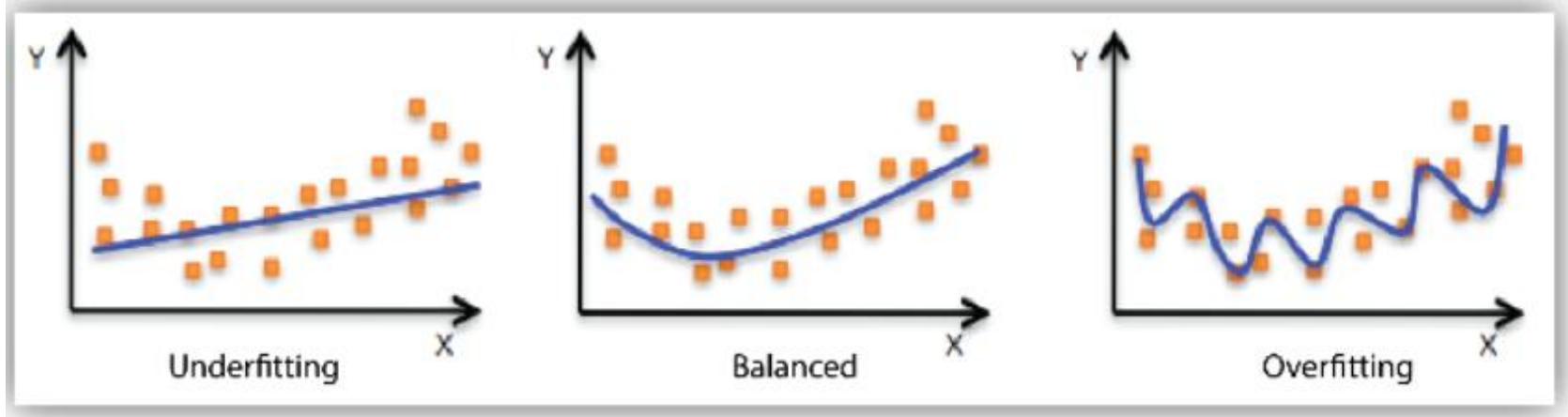
- Model learns too much about training data – on its fine details (noise)
 - Tailored shirt compared to a shirt from Uniqlo.

Overfitting

Can an overfitting model generalise in finance?



Over-fitting vs under-fitting



Bias vs Variance

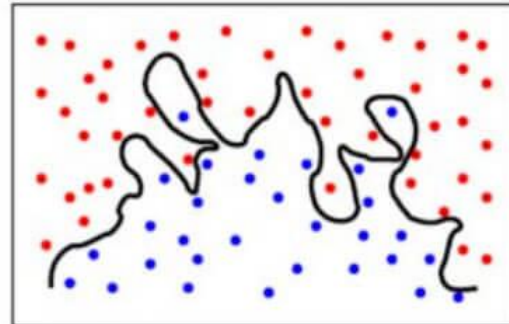
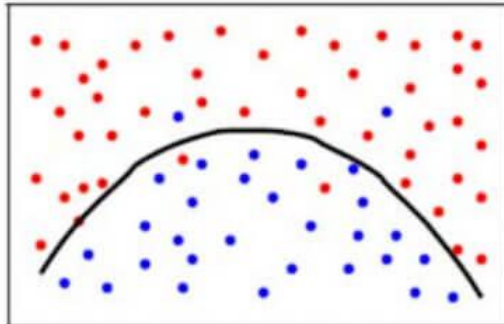
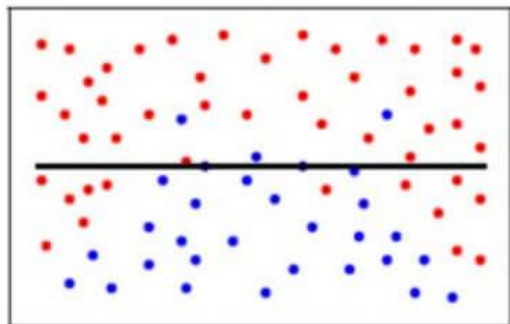
Overfitting vs. Underfitting

Generalization Problem in Classification

Underfitting



Overfitting

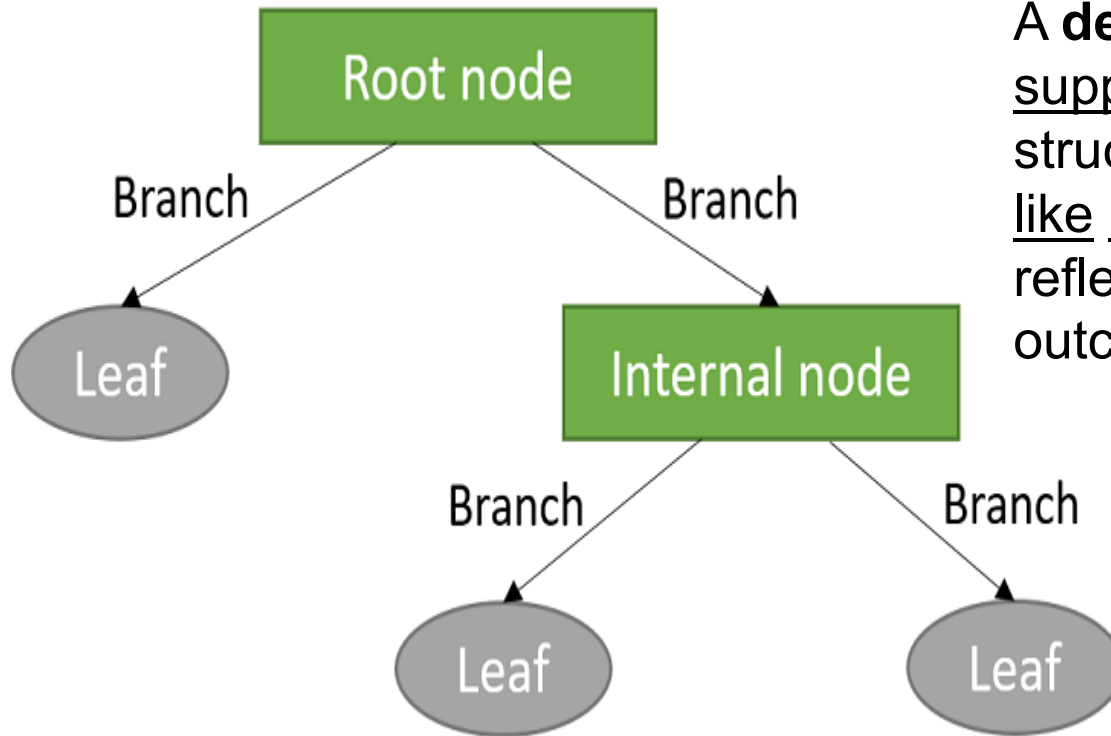


- Guess few times but get them right OR almost right every time?

Dive into (Some) Classification Models

- In practice, code to use different classification models in Python is 'replicable'.
- Using sci-kit library. Look at in example code:
 - Decision tree
 - Naïve Bayes
 - Logistic regression
- Different models have their own strengths or weaknesses (of the evaluation metrics).

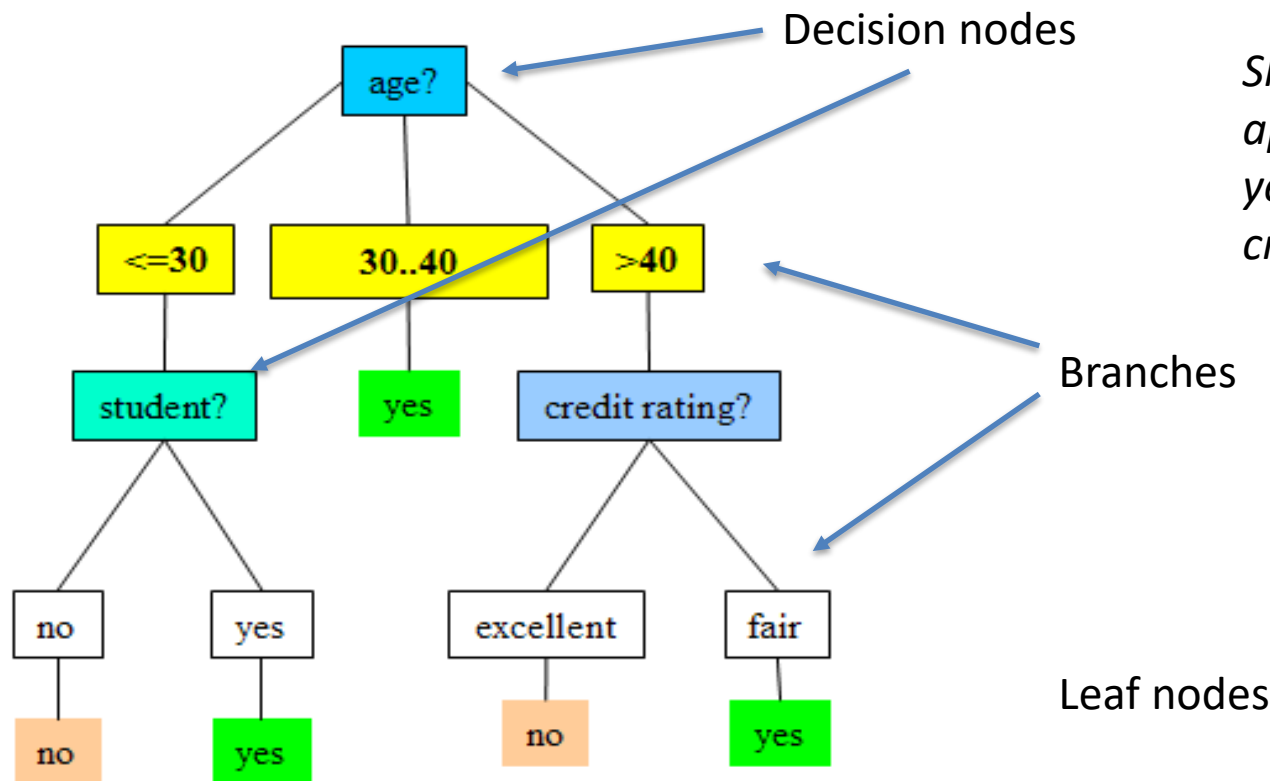
Defining a Decision Tree



A **decision tree** is a decision support recursive partitioning structure that uses a tree-like model of decisions reflecting their possible outcomes.

Using Decision Trees for loan approvals

- Transparent & easy to understand with series of 'rules'.
Should the loan application be rejected?



Should you reject a loan application for a man 45 year old with an excellent credit rating?

Strengths & Weaknesses

Strengths

- Easy-to-understand with a set of production rules.
- Applied to real problems.
- Make no prior assumptions about the data.
- Able to process both numerical and categorical data.
- Comparatively faster than Neural Networks.

Weaknesses

- Limited to one output attribute at a time
- Unstable (if one data changed, whole tree changes)
- Trees created from numeric datasets can be complex.

Naïve Bayes

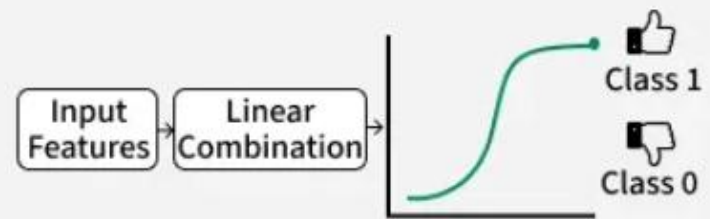
- Benchmark for ML method. Based on Bayes Rules – a statistical rule.
- Based on observed traits (eating kimchi), what is the likelihood of label 'X' (Korean)?
- For eg. About 10% of people eat kimchi. But almost ~95% of Koreans eat kimchi. So if we find someone who eats kimchi, can we say he is likely Korean?

Logistic Regression

- 1st generation technique. A statistical method with a linear combination of the predictor variables.

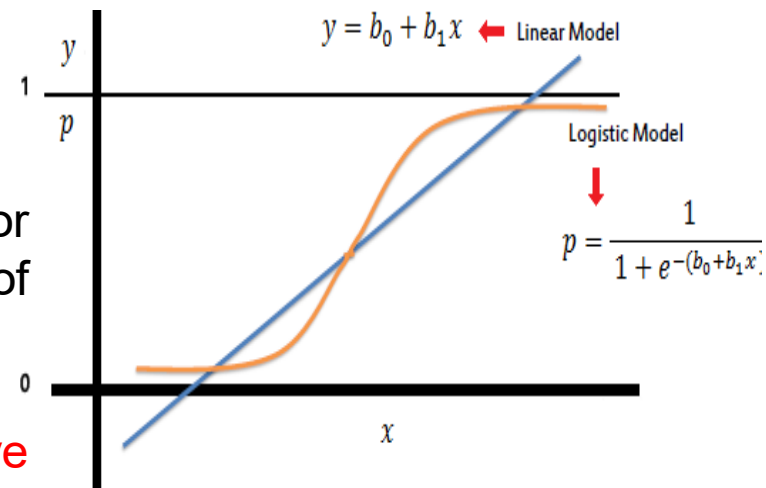
What is Logistic Regression?

- Predicts the probability of a binary outcome (Yes/No, 0/1)
- Uses the sigmoid function to map inputs to probabilities (0 to 1)
- Ideal for classification tasks



Assumptions of the Logistic Regression Model

- Logistic regression does **not assume a linear relationship** between the dependent and independent variables.
- The dependent variable have 2 categories.
- The independent variables need not be interval, nor normally distributed, nor linearly related, nor of equal variance within each group.
- The **categories (groups) must be mutually exclusive and exhaustive**; a case can only be in one group and every case must be a member of one of the groups.

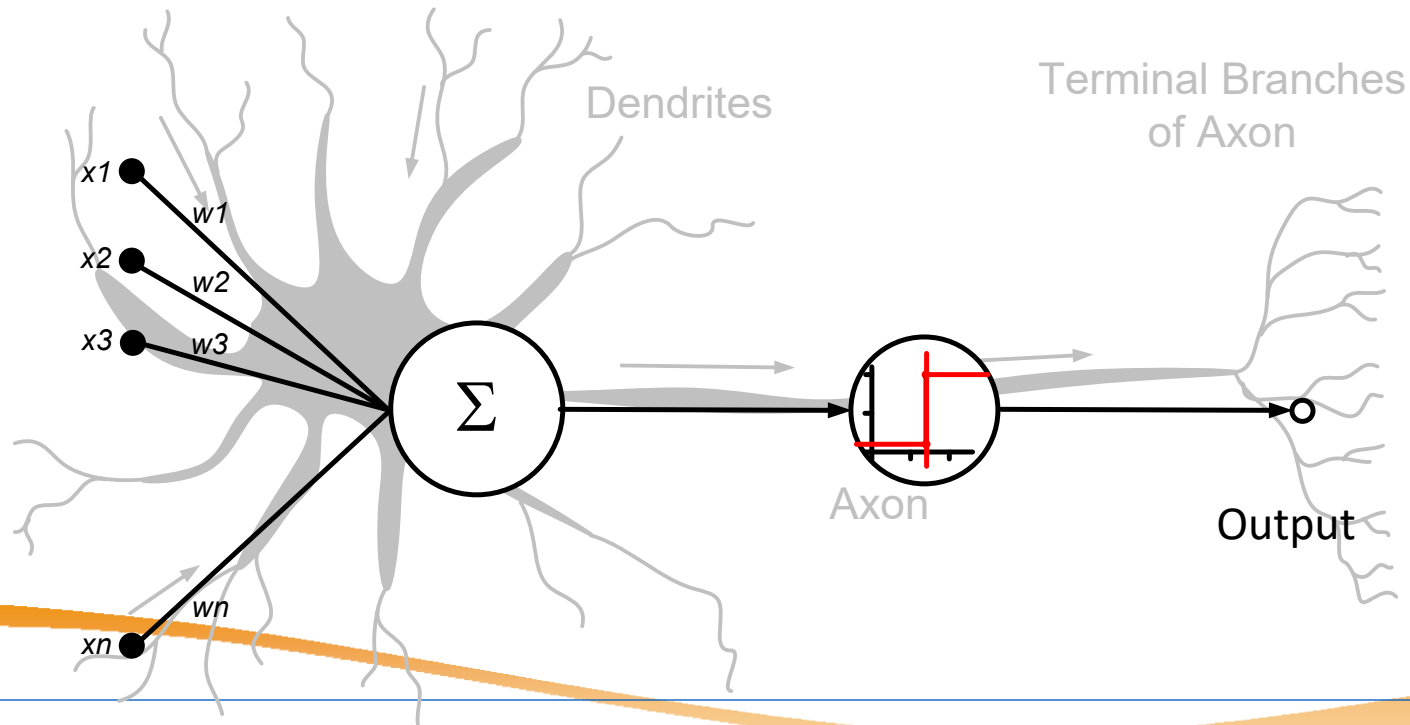


Basics of Underlying Theory

- For any set of predictive attributes, outcomes are assumed to be independently distributed binary (Bernoulli) random variables with probability of success p
 - p varies between 0 and 1 and is a function of predictor variables. The logistic regression provides the odds of being 1 to 0.
 - Odds = $(p/1-p)$
 - To give a classification, the estimated probability of being a 1 can be calculated based on a cut-off probability. For example, 0.5 can be used as a cut-off probability.
- 

Perceptron (Basic neural network)

- Basic building block of deep learning
 - 1960s : a neuron. Each x 's come with a weight (or parameter) to be adjusted
 - Threshold activation function



Deep Learning (DL) Network

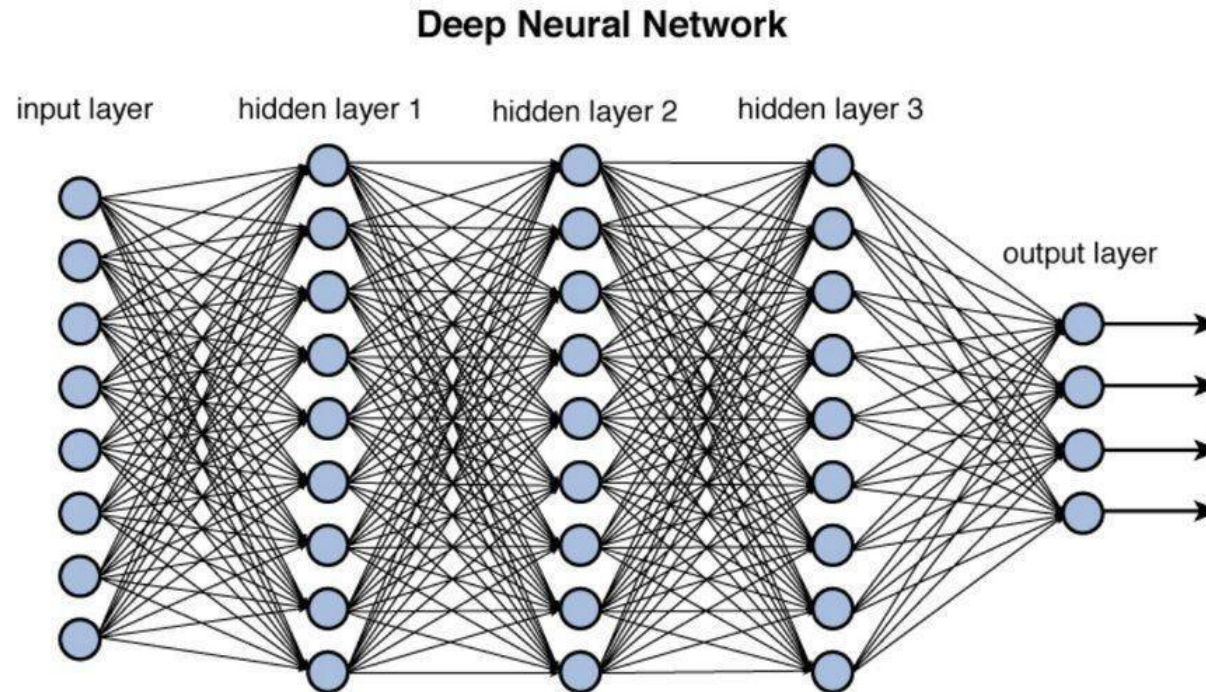
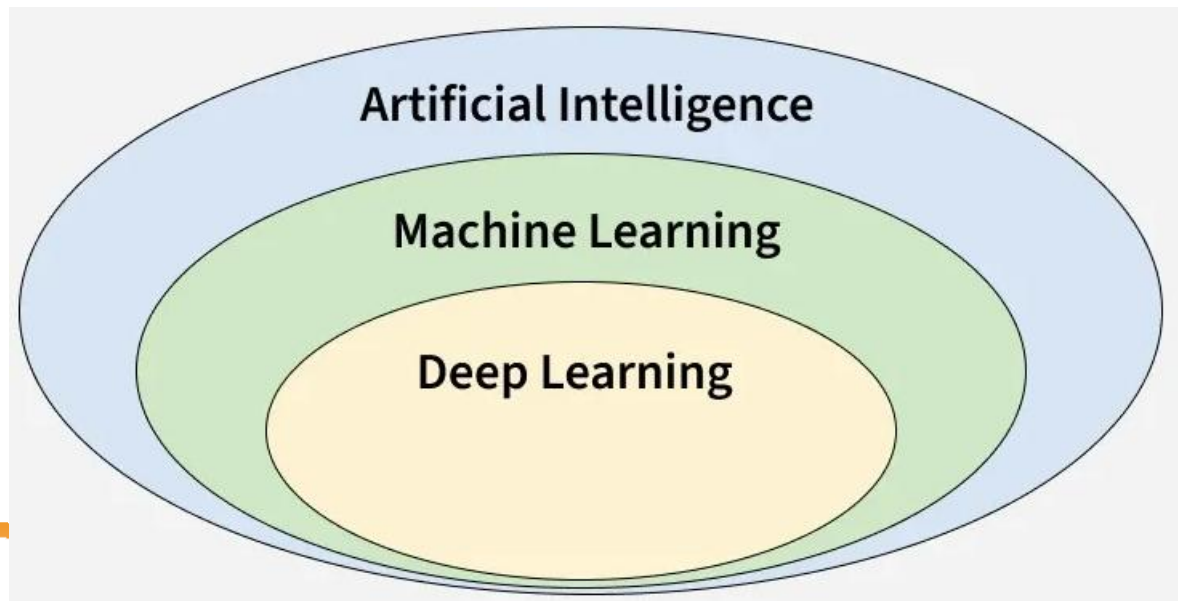


Figure 12.2 Deep network architecture with multiple layers.

- The neurons can be stacked horizontally and/or vertically to form a deep network. Finally an output.. *Huge number of weights*

Pros and Cons of DL Methods

No	Advantages	Disadvantages
1	Accurate	Takes a long time to train with high computational needs
2	Performs well on different types of and complex data	Could overfit due to large number of parameters
3		Black box problem



Use cases of Supervised learning methods

- Used for general prediction in classification and regression.
 - Loan applications
 - Sentiment Analysis
 - Trading algorithms
 - Customer churn prediction ...etc...

Can you think of any other?

Conclusion

- **ML Types in Finance:**

- *Supervised* (classification/regression with labels),
- *Unsupervised* (pattern discovery),
- *Reinforcement Learning* (agent–reward framework).

- **Supervised Learning Focus:**

- Logistic Regression,
- Naïve Bayes,
- Decision Trees,
- SVMs, Neural Networks

- **Evaluation Metrics:**

Confusion Matrix, Precision, Recall (TPR), ROC & AUC; trade-offs depend on **economic cost of errors** (no free lunch).

Conclusion

- **Key Risks:**

- Scarce & noisy financial data,
- non-stationarity,
- **overfitting vs underfitting** (bias–variance trade-off).

- **Model Insights:**

- Decision Trees = transparent & explainable;
- Naïve Bayes = simple benchmark;
- Logistic Regression = interpretable baseline;
- DL = powerful but prone to overfitting.

- **Finance Use Cases:**

Loan approval, fraud detection, customer prospecting, sentiment analysis, trading & asset management.

- **Core Takeaway:**

In finance, **generalisation and validation matter more than in-sample accuracy**—model choice and thresholds must align with business costs and explainability needs.

Case Study in Finance: Loan approval

- Scenario - Loan approval:
 - Suppose a bank built a model to approve application for credit loans. The bank is conservative in granting loans approval. The model identifies positive as defaulters and negative as non-defaulters.
 - Discuss which metrics are most important to evaluate models.
-
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Case Study in Finance: Customer Prospecting

- Scenario – Customer prospecting:
 - Suppose a model built for the marketing team to approach clients for prospective leads e.g. to use their services. The model does not want to ‘pester’ customers too much and identifies positive as those keen to engage their services.
 - Discuss which metrics are most important to evaluate models.
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Case Study: Funds Using ML Models

Background

In financial modelling, analysts often seek models that explain or predict returns, credit defaults, or frauds. With vast amounts of data—tick-by-tick prices, corporate financials, alternative datasets—it is tempting to fit highly complex models.

However, financial data is notoriously **noisy**, **non-stationary**, and subject to **regime shifts**. A model that fits historical data extremely well may fail in real markets. This is the essence of **overfitting**—capturing noise rather than signal.

Discussion: Using ML in Asset Management

Scenario

An asset management firm, **AlphaQuant**, designs a stock-selection model for its equity long/short fund.

Dataset: 15 years of daily data on S&P 500 constituents (prices, accounting ratios, analyst recommendations).

Target: Predict whether a stock will outperform the index over the next month.

Model chosen: A deep neural network with 12 hidden layers, trained on thousands of engineered features (momentum lags, valuation ratios, sentiment scores, etc.).

Performance in sample:

Training accuracy = **99%**

Backtest Sharpe ratio = **3.5**

Performance out of sample:

Test accuracy = **52%** (close to random guessing)

Sharpe ratio collapses to **0.2**

Discussion

What Happened?

Too many parameters vs. too little true signal: Finance rarely has strong stable patterns—models easily latch onto noise.

Data snooping bias: Repeated feature engineering and backtesting effectively mined spurious correlations (e.g., “stocks with tickers starting with M outperformed in 2014”).

Non-stationarity: Relationships that held in the past broke down as macro conditions and market structure evolved.

Lack of cross-validation: AlphaQuant optimised hyperparameters using the same period, inflating in-sample performance.

Discussion

Discussion Questions

- Why is overfitting particularly acute in financial datasets compared to, say, image recognition?
- How could AlphaQuant have improved its validation framework to avoid misleading backtests?
- What techniques exist to combat overfitting in finance?

You may use ChatGPT and the Internet for due diligence

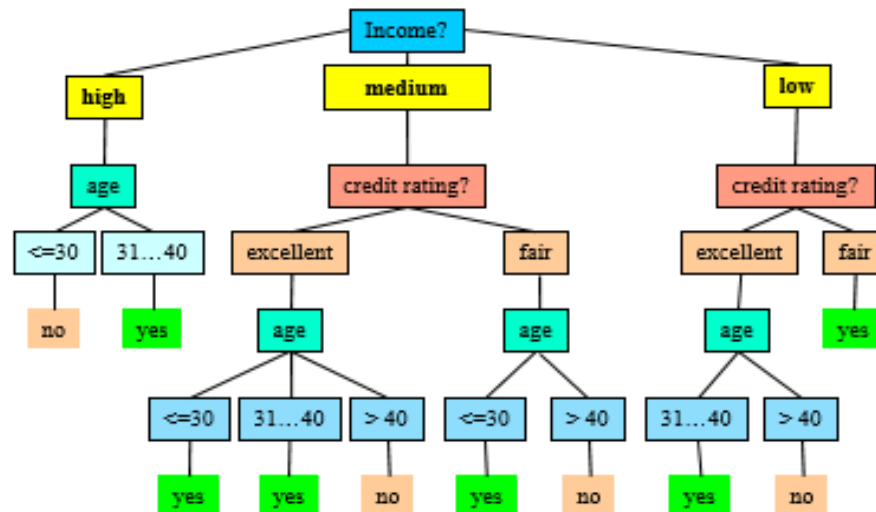
Quiz Time (together)



Quiz to test your understanding

1. A bank decides to use decision trees to approve or disapprove credit card applications. For someone who has an excellent credit rating and age 31-40, this application is most likely to be:

- a) Approved
- b) Not approved
- c) Cannot tell
- d) Need the income information as well



Quiz to test your understanding

Some (one) key disadvantage of using DL for prediction is:

- a. Tends to overfit
- b. Ability to model cross interaction effects
- c. Ability to model non-linear effects
- d. (b) and (c)



Quiz to test your understanding

Model Overfitting in Finance

Why is **overfitting** a particularly serious problem in financial machine learning models compared to image recognition?

- a) Financial data are stable and easy to model
- b) Financial data are noisy and non-stationary
- c) Markets always follow predictable cycles
- d) Overfitting does not affect financial models significantly



Quiz to test your understanding

The **Receiver Operating Characteristic (ROC)** curve plots which of the following pairs of metrics?

- a) Precision vs. Recall
- b) True Positive Rate vs. False Negative Rate
- c) True Positive Rate vs. False Positive Rate
- d) Sensitivity vs. Specificity

What is the use of the ROC?